Week 2 “Supersonic” by Demi Kanon

1. **Reference/Inspiration**:
   - **Song**: *"Supersonic"* by Demi Kanon
     - The song has a vibrant, energetic feel that can inspire creative coding projects involving rhythm and motion. The dynamic beats and melodic shifts could serve as a reference for creating generative art based on music or designing projects that respond to audio input in real-time.
   - Inspiration: Fusing electronic beats with an upbeat tempo is excellent for projects focused on sound-reactive visuals. This could guide work on generative visualizations where patterns or colours change in sync with the rhythm of the music.
2. **Creative Coding Project or Plan**:
   - Project Idea: Create an interactive generative art piece in which visuals respond to the beats of *"Supersonic*." This could involve using p5.js or Processing to create abstract visuals (like sound-reactive circles, waves, or particles) that pulse or morph based on the song’s rhythm and frequency analysis.
   - Challenges: Synchronizing audio input with visual patterns might be challenging, especially for real-time responsiveness. It may require techniques like Fourier Transform for audio spectrum analysis and mapping the results to visual elements.
3. **Tool, Technique, Library, or Tutorial to Explore**:
   - Tool: p5.js (or Processing)
     - It is an excellent library for creating generative art responding to music input. Specifically, p5.sound allows for audio analysis, which can be helpful in synchronizing visuals with sound. It will be essential for creating the generative, music-driven artwork you aim for.
   - Tutorial to Explore: Look into "Audio Visualization with p5.js" tutorials, which show how to analyze audio files (like *Supersonic*) to generate real-time visuals. You can find relevant resources on the p5.js website or platforms like YouTube or GitHub.

Week 3 Three.js

Reference/Inspiration:

- **Tool**: **Three.js**
    - Three.js is a powerful JavaScript library for creating 3D graphics in the browser. It simplifies working with WebGL, making it accessible for interactive, visually rich experiences. The inspiration comes from projects like the [Three.js Journey](), which showcases the library's capability for creating stunning generative art, immersive 3D environments, and interactive web-based visualizations. The ability to render 3D objects, incorporate real-time physics, and integrate dynamic lighting makes it an excellent tool for generative art and interactive applications.

Creative Coding Project or Plan:

- **Project Idea**: Build an interactive 3D web portfolio using Three.js.
    - The portfolio will showcase my projects in a visually engaging 3D environment. Users can navigate a dynamic scene to explore my work.
- **Features to Include**:
    - A central 3D space (like a gallery or futuristic room) with clickable 3D objects representing different projects.
    - Each object, when clicked, displays project details, links to GitHub, and visuals.
    - Interactive camera movements and transitions to make navigation smooth and engaging.
        - Use dynamic lighting and animations to bring the scene to life.
- **Tools & Techniques**:
    - Three.js for rendering the 3D environment.
    - GSAP for smooth animations and transitions between scenes or project details.
    - HTML/CSS to handle UI elements like menus or additional information.
    - Textures and Models: Use tools like Blender or free 3D asset libraries for custom objects to represent projects.
- **Challenges**:
    - Designing a user-friendly navigation system for the 3D space.
    - Optimizing performance for smooth interactions on various devices.
    - Managing the balance between interactivity and usability—ensuring the portfolio remains functional and not just a visual experience.

Week 4 Art from Code

Reference/Inspiration: https://art-from-code.netlify.app/day-1/session-1/

Tool: Art from Code - Day 1, Session 1
The tutorial introduces the fundamentals of generative art using code. It explores how simple loops, randomness, and basic shapes can create dynamic and visually interesting patterns. This session is a great starting point for understanding how to generate art programmatically.

Creative Coding Project or Plan: Plotting Data with Generative Art

- Use p5.js to create a visual representation of plotted data.
- Plot the data in a way that generates patterns and art.
- Experiment with different ways to map data points onto the canvas, using shapes, colours, and patterns inspired by generative art techniques.
- Implement randomness and interactivity to make the visualization more engaging.

Tools & Techniques:

- p5.js for drawing and rendering graphics.
- Loops and randomness to generate patterns dynamically.
- Data mapping to position elements based on numerical values.

Challenges:

- Finding a balance between randomness and structured data representation.
- Ensuring the visualization effectively communicates the data while still being aesthetically interesting.

Week 5 Boid Flocking

Reference/Inspiration: Concept: Boid Flocking Algorithm https://eater.net/boids

Boid Flocking is a simple rule-based simulation of collective animal behaviour, such as bird flocks or fish schools. It is inspired by Craig Reynolds' 1986 "Boids" model, which demonstrates how local agent interactions can lead to complex, lifelike movement patterns. This concept is widely used in computer graphics, simulations, and generative art.

Creative Coding Project or Plan:

- Implement a **boid simulation** where particles (boids) move according to three primary rules:
  - **Separation:** Avoid crowding neighbours.
  - **Alignment:** Steer towards the average direction of nearby boids.
  - **Cohesion:** Move toward the center of the flock.
- Experiment with parameters like speed, field of view, and interaction strength.
- Add **user interaction**, allowing control over boid behaviour (e.g., clicking to attract or repel boids).

Tools & Techniques:

- **p5.js** for rendering and animation.
- **Vector mathematics** for steering behaviours.
- **Loops and conditionals** to handle interactions between multiple boids.
- **Mouse or keyboard input** to modify flocking behaviour dynamically.

Challenges:

- Balancing movement rules for realistic flocking behaviour.
- Optimizing performance when handling many boids at once.
- Fine-tuning parameters to create aesthetically pleasing motion.

Week 6

Week 7

Week 8